Shivansh Kalra

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Education

- **Shiv Nadar University** — Greater Noida, India
  Bachelor of Technology in Computer Science — *Aug. 2023 – May 2027 (Expected)*
  - Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming

Technical Skills

Languages: Python, JavaScript/TypeScript, C++, Node.js, SQL, Bash, Rust (learning)
Frameworks & Systems: React, Next.js, NestJS, FastAPI, Docker, Redis, Microservices, REST APIs
AI / ML: LLM integration, RAG systems, RL environments (OpenAI Gym/OpenEnv), GARCH/HMM, ChromaDB
Databases & Infra: PostgreSQL, MongoDB, Firebase, AWS (EC2, S3, Lambda), GitHub Actions CI/CD
Core CS: Algorithms & Data Structures, MVCC internals, Static Analysis, Concurrency, Systems Programming

Experience

- **Pactle** — Remote
  Founding Software Engineer — *Feb. 2026 – Present*
  - Building the generalized product from the ground up, owning core technical foundations and scalable system architecture

- **Thinkly Labs** — Remote
  Full Stack Engineer — *Nov. 2025 – Jan. 2026*
  - Built North, a content generation engine from scratch — owned DB schema, API design, and frontend end-to-end; collaborated with founding team on distributed system architecture decisions

- **FilFlo** — Remote
  Backend Engineer — *Sept. 2025 – Nov. 2025*
  - Architected microservices-based backend using Node.js and NestJS, migrating 50% of monolithic codebase to distributed architecture and reducing deployment time by 40%
  - Implemented Redis caching layer, cutting response latency from 800ms to sub-300ms (37% improvement) for 10,000+ daily requests while maintaining 99.8% uptime; introduced Docker containerization across environments

- **Testline** — Remote
  Product Development Intern — *Dec. 2024 – Feb. 2025*
  - Delivered 3 production features serving 3,000+ active users; engineered authentication system reducing signup friction by 11%
  - Achieved 85% code coverage with unit and integration tests, proactively resolving 25+ critical bugs before deployment

- **Frontrunner** — Remote
  Software Development Intern — *May 2024 – Aug. 2024*
  - Shipped 2 full-stack applications (React, Node.js, MongoDB) within 8-week sprints; built CI/CD pipelines with GitHub Actions reducing manual deployment effort by 70%
  - Collaborated with 5+ engineers via Agile standups and code reviews to deliver scalable web applications

- **CatalystIQ** — Gurgaon, India
  Founder & Technical Lead — *Jan. 2024 – 2025*
  - Architected enterprise-grade AI platform (React, Node.js, Python, PostgreSQL) serving 10+ B2B clients at 99%+ uptime; ML-based recruitment module reduced client hiring cycle time by 65%

Projects

- **InVariant** | *Python, TypeScript, Rust, Static Analysis, Playwright*
  - Multi-component security platform: Python static analysis engine (90%+ report accuracy), autonomous LLM reasoning agent using Playwright for live system auditing, Next.js web app, and Rust CLI for cloud infrastructure inspection
  - Authenticated agent operates as a self-directed reasoning loop — plans and executes multi-step browser actions combining static signals with dynamic runtime observation; shipping weekly iterations
- **Synthetic Market RL Environment (open-env-hack)** | *Python, FastAPI, OpenAI Gym/OpenEnv, Docker*
  - Built synthetic financial market environment for LLM agent evaluation; engine uses GARCH(1,1) volatility clustering and HMM regime switching — simulates dynamics rather than replaying data, making lookahead bias structurally impossible
  - Three difficulty tiers with Poisson news-shock injection and Cholesky correlation sampling; deployed via FastAPI + Docker with full validation suite
- **PACER — MVCC Garbage Collection** | *Systems, Database Internals, Concurrency*
  - Designed novel pressure-adaptive, lock-free, continuous GC algorithm for MVCC databases with snapshot coalescing to minimize write amplification under high-concurrency workloads; published as academic paper
- SNUXplore | *Next.js, MongoDB, Azure AI Services, Microservices*
  - Scaled to 30,000+ concurrent users via distributed architecture with load balancing, multi-layer caching, and query optimization achieving sub-100ms response times; integrated RAG-based LLM for natural-language query resolution

Research & Achievements

- **PIE: Physics-Inspired Contrastive Learning for Low-Light Image Enhancement** | *IEEE-formatted*
  - Bag-of-Curves negative sample generation, Gram matrix triplet loss + InfoNCE objective, unsupervised superpixel segmentation; outperformed RetinexNet, Zero-DCE, and EnlightenGAN on the LOL dataset
- **Prediction Markets as Macro Derivatives** | *Finverse 2026 Research Competition*
  - Modelled Kalshi binary event contracts as macro derivatives using modified Black-Scholes on VIX surface across 383 contracts; hedged portfolio achieved Sharpe 0.538 vs. unhedged 0.065 at 1% NAV allocation

Top 3 finishes in 5+ hackathons across AI/ML and blockchain domains | Selected participant, Mercer Summit 2025 — Marsh & McLennan | Strong competitive programming foundation in DSA